Jorge Sánchez Aguilar

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SUMMARY

Final-year CS student and AI Engineer at Ferrovial Digital Hub building production computer vision systems for US highway infrastructure, and NASA Space Apps 2025 Global Finalist. Focus: end-to-end AI systems from model training to deployment, and multi-agent LLM architectures.

EXPERIENCE

AI Engineer Intern — Digital Hub, Ferrovial — Sept 2025 – Present

Madrid, Spain (Hybrid) · Computer Vision team — US Highway Tolling Projects

- Own the end-to-end computer vision pipeline: model training, data annotation, evaluation, containerized deployment, and automated reporting for business stakeholders.
- Built a night-time vehicle detection model from scratch using YOLO26-Pose for low-quality public CCTV cameras with heavy headlight noise; selected over CenterNet and YOLOv8 alternatives developed in parallel, now the production candidate.
- Designed and shipped a non-AI camera perspective-shift detector using geometric heuristics and a 3-check validation gate (2-of-3 required), detecting sub-pixel CCTV movement in real time; selected for production over ML alternatives due to speed and accuracy.
- Stack: Python, PyTorch, OpenCV, Docker, Azure DevOps, Azure Blob Storage.

PROJECTS

Multi-Agent LLM System for Financial Market Analysis — March 2026 – Present

Solo project — private repository

- Designed a reasoning-first multi-agent pipeline where specialized LLM agents collaborate on market analysis: from macro scanning through per-asset analysis to portfolio-level risk review.
- Made deliberate engineering tradeoffs on LLM vs. deterministic logic: agents handle reasoning and analysis, while portfolio-level constraints remain as deterministic Python for reliability.
- Deployed local inference with Qwen3.5-35B-A3B on vLLM (RTX 5090, 32GB VRAM) with tuned GPU memory utilization; maintained OpenRouter fallback for API-based experimentation.

Space Atlas — NASA Space Apps Challenge — Oct 2025

Team of 5 · Madrid Winner · Global Finalist

- Owned AI integration and cloud deployment: embedded a GPT-4 Vision chatbot into a full-stack 3D solar system web platform for exploring NASA high-resolution planetary imagery.
- Architected and deployed the full Azure stack (App Services for frontend and backend, custom domain, production environment) end-to-end during the hackathon weekend.

SnakeAI — Deep Reinforcement Learning — Jul 2025

Solo project

- DQN-based agent for Snake with custom reward shaping and GPU-accelerated PyTorch training; includes interactive gameplay interface for real-time testing.

EDUCATION

B.S. Computer Science — Universidad Rey Juan Carlos — Expected July 2026

Concentrations: Artificial Intelligence, Modeling & Simulation. Madrid, Spain.

42 Madrid — Peer-to-peer software engineering program — 2023 – 2025

Level 3 (80% complete). Focus: C/C++ systems programming, UNIX/Linux, object-oriented design, AI/ML.

AWARDS & COMPETITIONS

- **NASA Space Apps Challenge 2025 — Madrid Winner & Global Finalist**. Advanced to global finalist round out of 11,500+ teams and 114,000+ participants worldwide.
- **Qubic × Vottun Madrid Hackathon 2025 — Top 10 Finalist**

SKILLS

AI / ML: Computer Vision, Object Detection, Pose Estimation, Deep Reinforcement Learning, LLMs, Multi-Agent Systems, Prompt Engineering, GenAI, vLLM.

Programming: Python (PyTorch, NumPy, Pandas, OpenCV), C, C++, Java, Scala, SQL.

Tools & Cloud: Docker, Git, Azure (OpenAI, Blob Storage, App Services, DevOps), Linux/UNIX.

Languages: English (C2 Proficiency), Spanish (Native).